

⑦ критерий Дарбу

$$\textcircled{8} \quad \lim_{\lambda(\tau) \rightarrow 0} (S_\tau(f) - s_\tau(f)) = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} S_\tau(f) - \text{верхняя} \\ s_\tau(f) - \text{нижняя} \end{cases} \Rightarrow \begin{cases} S_\tau(f) - \text{ср} \\ s_\tau(f) - \text{ср} \end{cases}$$

$$\Rightarrow \exists \sup_{\tau} s_\tau(f) = I_* < +\infty$$

$$\exists \inf_{\tau} S_\tau(f) = I^* < +\infty$$

$$I_* \leq I^*$$

$$\forall \tau: S_\tau(f) \leq I_* \leq I^* \leq S_\tau(f)$$

$$\Rightarrow 0 \leq I^* - I_* \leq S_\tau(f) - s_\tau(f) < \varepsilon$$

$$\xrightarrow{\lambda(\tau) \rightarrow 0} 0 \quad \xrightarrow{\lambda(\tau) \rightarrow 0} 0$$

$$\Rightarrow I^* = I_* \quad \exists I = I_* = I^*$$

$$\Rightarrow \text{т.к.} \int s_\tau(f) \leq I \leq S_\tau(f)$$

$$\int s_\tau(f) \leq \mathcal{D}_\tau(f, \xi) \leq S_\tau(f)$$

$$\Rightarrow |\mathcal{D}_\tau(f, \xi) - I| \leq S_\tau(f) - s_\tau(f) < \varepsilon$$

$$\stackrel{\text{def}}{=} \Rightarrow I = \int_a^b f(x) dx \quad \triangle$$

Опр $I_* = \sup_{\tau} s_\tau(f)$ наг. числом Дарбу

$$I^* = \inf_{\tau} S_{\tau}(f) \text{ на } \underline{\text{сегменте}} \text{ непрерывн}$$

Дарф

Итак если $\exists I_*, I^* \text{ и } I_* = I^*, \text{ то}$

$$\exists I = \int_a^b f dx = I_* = I^*$$

Зам на $\textcircled{I}$ Дарф $\Rightarrow$

$$S_{\varepsilon}(f) \leq \int_a^b f dx \leq S_{\tau}(f) \quad \textcircled{*}$$

Следствие Если $f \in R[a, b] \Rightarrow$

$$\lim_{\lambda(\tau) \rightarrow 0} S_{\tau}(f) = \lim_{\lambda(\tau) \rightarrow 0} S_{\tau}(f) = \int_a^b f dx$$

$$\Delta \textcircled{*} \Rightarrow \begin{cases} 0 \leq S_{\tau}(f) - \int_a^b f dx \leq S_{\tau}(f) - S_{\varepsilon}(f) < \varepsilon \\ 0 \leq \int_a^b f dx - s_{\tau}(f) \leq S_{\tau}(f) - s_{\varepsilon}(f) < \varepsilon \end{cases}$$

Кр Дарф $\Rightarrow$

$$\begin{cases} |S_{\tau}(f) - \int_a^b f dx| < \varepsilon \\ |s_{\tau}(f) - \int_a^b f dx| < \varepsilon \end{cases} \Rightarrow \begin{cases} \lim_{\lambda(\tau) \rightarrow 0} S_{\tau}(f) = \int_a^b f dx \\ \lim_{\lambda(\tau) \rightarrow 0} s_{\tau}(f) = \int_a^b f dx \end{cases}$$

Δ

Опр $f: E \rightarrow \mathbb{R}$. Кусочно
 ϕ -сум f на E наг. беспрер

$$\omega(f, E) = \sup_{x, y \in E} (f(x) - f(y))$$

$$\text{или } \omega(f, E) = \sup_{x \in E} f(x) - \inf_{x \in E} f(x)$$

Существование (кр. Дарбу в термине $\omega(f, \xi)$)
 $f \in R[a, b] \Leftrightarrow \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \omega(f, \Delta_i) \cdot \Delta x_i = 0$

т.е. $\forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0 : \forall \tau : \lambda(\tau) < \delta$
 $\Rightarrow \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i < \varepsilon$

Зам.

$$\begin{aligned} (S_\tau(f) - s_\tau(f)) &= \sum_{i=1}^n M_i \Delta x_i - \sum_{i=1}^n m_i \Delta x_i = \\ &= \sum_{i=1}^n \sup_{x \in \Delta_i} f(x) \Delta x_i - \sum_{i=1}^n \inf_{x \in \Delta_i} f(x) \Delta x_i = \\ &= \sum_{i=1}^n \left(\sup_{x \in \Delta_i} f(x) - \inf_{x \in \Delta_i} f(x) \right) \Delta x_i = \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i \end{aligned}$$

Ⓘ Если $f \in C^0[a, b] \Rightarrow f \in R[a, b]$

Δ] $f \in C^0[a, b] \xRightarrow{\text{Т Кантора}} f$ — равномерно
 непрерывна на $[a, b] \Rightarrow \forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0 :$

$\forall x_1, x_2 \in [a, b] : |x_1 - x_2| < \delta \Rightarrow$

$$|f(x_1) - f(x_2)| < \varepsilon$$

] $\tau : \lambda(\tau) < \delta \Rightarrow$

$$\omega(f, \Delta_i) = \sup_{x, y \in \Delta_i} |f(x) - f(y)| < \frac{\varepsilon}{b-a}$$

$$\Rightarrow \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i < \frac{\varepsilon}{b-a} \sum_{i=1}^n \Delta x_i = \varepsilon$$

и.к.р. $\Rightarrow f \in R[a, b]$

① Если значение интеграла Φ — чин
 измелить не конечен или не годек,
 то интеграл Φ — чин не каруинте,
 и интеграл не измелится

Δ] $f \in R[a, b]$,] $\tilde{f}$ отим-се в гг:
 $x_1, x_2, \dots, x_n$. $\Rightarrow$ f -ср. на $[a, b] \Rightarrow$

$$\exists C > 0: |f| \leq C \Rightarrow |\tilde{f}| \leq \tilde{C}$$

$$\tilde{C} = \max(C, |f(x_1)|, |f(x_n)|) \Rightarrow$$

$$\exists |\delta_\varepsilon(f, \xi) - \delta_\varepsilon(\tilde{f}, \xi)| \leq 2(nC + n\tilde{C})\lambda(\varepsilon) \xrightarrow{\lambda(\varepsilon) \rightarrow 0} 0$$

$$\Rightarrow I = \lim_{\lambda(\varepsilon) \rightarrow 0} \delta_\varepsilon(f, \xi) = \lim_{\lambda(\varepsilon) \rightarrow 0} \delta_\varepsilon(\tilde{f}, \xi) \Delta$$

$$|A - B| = |A + (-B)| \leq |A| + |B|$$

① ① Если $f \in R[a, b]$ и $[\alpha, \beta] \subset [a, b]$
 $\Rightarrow f \in R[\alpha, \beta]$

② Если $\int f \in R[a, c]$ и $\int f \in R[c, b]$ $a < c < b \Rightarrow$
 $\int f \in R[a, b]$

Δ 1)] $f \in R[a, b]$ $\stackrel{\text{кр. D}}{\Rightarrow} \forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0:$

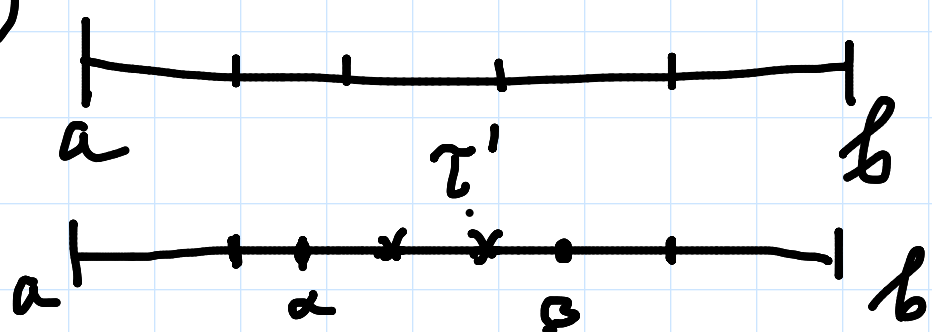
$$\forall (\tau, \xi): \lambda(\tau) < \delta \Rightarrow \underline{S_\tau(f)} - \underline{S_\tau(f)} < \varepsilon$$

$$\exists \tau': [\alpha, \beta], \lambda(\tau) < \delta$$

$$\lambda(\tau) < \delta$$

$$< \varepsilon$$

②



$$0 \leq \underline{S_{\tau'}(f)} - \underline{S_{\tau'}(f)} \leq \underline{S_\tau(f)} - \underline{S_\tau(f)} < \varepsilon$$

$$\stackrel{\text{KP}^2}{=} \quad f(x) \in R [a, b] \quad \Delta$$

$$2) \quad \exists f \neq \text{const} \Rightarrow \omega(f, [a, b]) > 0$$

$$\forall \varepsilon > 0 \quad \left\{ \begin{array}{l} \exists \delta_1 = \delta_1(\varepsilon) > 0 : \forall (\tau_1, \xi_1) : \lambda(\tau_1) < \delta_1 \\ \exists \delta_2 = \delta_2(\varepsilon) > 0 : \forall (\tau_2, \xi_2) : \lambda(\tau_2) < \delta_2 \end{array} \right. \quad \stackrel{[a, c]}{\Rightarrow}$$

$$\Rightarrow \left\{ \begin{array}{l} S_{\tau_1} - s_{\xi_1} < \varepsilon/3 \\ S_{\tau_2} - s_{\xi_2} < \varepsilon/3 \end{array} \right. \quad \exists \delta = \min(\delta_1, \delta_2, \frac{\varepsilon}{\omega(f, [a, b])})$$

$$\exists \tau_{[a, b]} : \lambda(\tau) < \delta, \quad \exists C \in [x_{i-1}, x_i];$$

$$\text{Def-}u: \quad \tau' = \tau \cup C, \quad \tau_1 = \tau' \cap [a, c] \\ \tau_2 = \tau' \cap [c, b]$$

$$\begin{aligned} \text{a) } \quad \underbrace{S_{\tau'}(f) - s_{\xi}(f)}_{< \varepsilon/3} &\leq S_{\tau_1}(f) - s_{\tau_1}(f) + \\ &+ S_{\tau_2}(f) \stackrel{\leq \varepsilon/3}{\rightarrow} s_{\tau_2}(f) + \omega(f, [a, b]) \delta < \varepsilon \quad \lambda \\ &\Rightarrow f \in R[a, b] \quad \Delta \end{aligned}$$